CBOT WHEAT FUTURES (ZW)

Soft Red Winter Wheat — Spread & Curve Structure Fundamental Reference

The Multi-Class, Multi-Hemisphere Global Wheat Market and the CBOT Benchmark

This document presents a fact-based reference on the fundamental drivers of CBOT Wheat futures (ticker: ZW), contract size 5,000 bushels, pricing Soft Red Winter (SRW) wheat — the longest-established and most heavily traded of the several US wheat futures contracts and the global benchmark price reference most frequently cited in general financial media coverage of "wheat prices." Wheat is structurally distinct from corn (covered in a separate reference document in this series) in several important respects: it is grown across multiple classes with materially different end-use applications, it is planted and harvested on two distinct calendars within the US alone (winter wheat and spring wheat), and — most importantly for a global price-discovery perspective — it is grown across a far more dispersed set of major producing countries spanning every populated continent and, critically, multiple hemispheres with offsetting harvest calendars, such that there is effectively always an active wheat harvest somewhere in the world. This document presents the SRW/CBOT-specific fundamentals in full, including the wheat-class structure that distinguishes CBOT wheat from the Kansas City (hard red winter) and Minneapolis (hard red spring) wheat contracts, the global, multi-hemisphere production calendar, the Black Sea region's structural importance to global wheat trade, and the cross-market relationship to corn that governs feed-wheat substitution economics.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Delivery Mechanics

CBOT Wheat prices Soft Red Winter (SRW) wheat, one of several distinct wheat classes grown in the US, each suited to different milling and baking applications and each, in the US futures market, priced via its own dedicated contract on a different exchange.

Parameter	Detail
Exchange	CME Group — Chicago Board of Trade (CBOT)
Ticker	ZW (front month)
Underlying class	Soft Red Winter (SRW) wheat
Contract size	5,000 bushels (approximately 136 metric tonnes)
Quotation	US cents per bushel (¢/bu); minimum tick = 1/4 cent/bu = $12.50 per contract
Contract months	March (H), May (K), July (N), September (U), December (Z) — five delivery months per year. July is the primary new-crop (harvest) contract, reflecting winter wheat's early-summer US harvest — a different new-crop timing from corn's December anchor (see Section 1.1).
Settlement type	Physical delivery via shipping certificate at CBOT-approved grain elevators and shipping stations in the Chicago/Illinois River and Toledo, Ohio delivery markets.
Delivery grade	No. 2 Soft Red Winter at par, with other grades and classes (including some hard wheat classes) deliverable at exchange-set premiums and discounts, per USDA grading standards.
Trading hours	CME Globex: Sunday–Friday 19:00–07:45 ET and 08:30–13:20 ET, the same general electronic-session structure as corn.
Last trading day	Business day prior to the 15th calendar day of the delivery month
Contract value at $6.00/bu	$30,000 per contract
US wheat marketing year	June 1 – May 31, aligned with the US winter wheat harvest — a different marketing-year start date from corn's September 1 convention.

1.1 Why the New-Crop Anchor Contract Differs From Corn

- Because US winter wheat (the dominant US wheat type by volume, including SRW) is planted in the autumn, overwinters as an established plant, and is harvested in late spring/early summer (roughly May through July depending on latitude), the new-crop anchor contract for CBOT wheat is July — the contract referencing the crop just harvested — in contrast to corn's December anchor, which reflects corn's spring-planted, autumn-harvested single annual cycle. This difference in harvest timing is the most basic structural distinction between the two grains' calendars and a recurring source of differing seasonal spread behaviour between the two markets.

1.2 Old-Crop / New-Crop Structure

- As with corn and cotton, the old-crop (March, May) versus new-crop (July, September, December) distinction organises the wheat curve: March and May price wheat from the harvest before last that remains in storage and is being marketed through the June–May marketing year, while July prices the crop just harvested. The May-July spread (rather than corn's July-December spread) is therefore the primary old-crop/new-crop transition spread for CBOT wheat, reflecting the different harvest timing described in Section 1.1.

2. Wheat Classes — Why Multiple US Wheat Contracts Exist

Unlike corn, where a single class (yellow dent corn) dominates US production and the CBOT contract, wheat is grown in the US (and globally) across several distinct classes with materially different protein content, hardness, and milling/baking characteristics — differences significant enough that the US futures market prices the major classes via entirely separate contracts on different exchanges rather than a single unified contract.

Class	Futures Contract	Primary Growing Region	Protein/Hardness	Primary End Use
Soft Red Winter (SRW)	CBOT Wheat (ZW) — this document	Eastern Corn Belt, Ohio Valley, parts of the Mid-South and Southeast (Ohio, Indiana, Illinois, Missouri, Arkansas, and others)	Lower protein, soft endosperm	Flat breads, crackers, cakes, pastries, and other products requiring a softer, lower-protein flour; also a significant feed-wheat substitute for corn when relative prices favour it (Section 7).
Hard Red Winter (HRW)	Kansas City Wheat (KE), traded on CBOT/CME after the KCBT merger	Central and Southern Great Plains (Kansas, Oklahoma, Texas Panhandle, Nebraska, Colorado)	Higher protein, hard endosperm	All-purpose and bread flour; the largest single US wheat class by volume and the most significant US wheat export class.
Hard Red Spring (HRS)	Minneapolis Wheat (MWE), traded on MGEX/CME	Northern Great Plains (North Dakota, Minnesota, Montana, South Dakota)	Highest protein among major US classes	Premium bread flour and applications requiring the highest protein/gluten strength; often carries a quality premium over winter wheat classes.
Soft White	No major dedicated CBOT-family futures contract; priced via cash/basis markets referencing SRW	Pacific Northwest (Washington, Oregon, Idaho)	Low protein, soft endosperm	Noodles, flat breads; a significant US export class, particularly to Asian milling markets.
Durum	No major actively-traded US futures contract	Northern Plains (North Dakota primarily)	Very hard endosperm, high protein	Pasta production almost exclusively.

2.1 The SRW/CBOT Contract's Specific Role

- Because SRW is not the largest US wheat class by production volume (Hard Red Winter holds that position), the CBOT contract is, in a strict physical-delivery sense, a regional benchmark for the Eastern Corn Belt/Ohio Valley growing area specifically. Nonetheless, CBOT Wheat is, by a wide margin, the most actively traded and most liquid of the US wheat futures contracts, and has historically functioned as the default global "wheat price" reference in general financial commentary and as the most commonly used instrument for broad wheat-market hedging and speculative exposure, even by participants whose underlying physical exposure may be to a different wheat class entirely.
- **The wheat-class spread complex**: because KC (HRW) and Minneapolis (HRS) wheat trade as separate, exchange-listed contracts, the spreads between CBOT (SRW), KC (HRW), and Minneapolis (HRS) wheat are themselves tracked and, at times, traded relative-value relationships, reflecting relative regional crop conditions, protein premiums, and class-specific export demand — broadly analogous in concept (though arising from class/quality differences rather than currency or processing-stage differences) to the cross-contract comparisons described for cocoa, sugar, and

coffee elsewhere in this document series.

3. The US Winter Wheat Crop Calendar

Winter wheat's autumn-planting, overwintering, and late-spring/early-summer harvest cycle is fundamentally different from corn's single spring-to-autumn cycle, introducing weather risk windows (autumn planting/establishment and winter dormancy) that have no equivalent in the corn calendar.

Period	Stage	Key Weather Risk	Primary Data Release	ZW Curve Impact
Sep–Oct	Planting and emergence	Dry soils delaying emergence and stand establishment ahead of winter dormancy; poor fall establishment increases winterkill risk	USDA weekly Crop Progress (planted/emerged %)	Establishes the baseline condition of the crop entering dormancy; poor fall conditions are an early, though not yet decisive, bearish-for-yield signal carried into the following spring
Nov–Feb	Winter dormancy	Winterkill — extreme cold without adequate snow cover insulation can kill the dormant plant outright; this is a genuine, if less dramatic, parallel to the frost-kill risk described for Brazilian coffee, though affecting a dormant plant rather than an actively fruiting tree	USDA does not publish weekly Crop Progress during dormancy; satellite-based vegetation health indices and state extension service reports are the primary monitoring tools during this window	Generally low day-to-day market attention during dormancy itself, but a severe cold event without snow cover can produce a significant condition-rating revision once weekly reporting resumes in spring
Mar–Apr	Spring green-up and resumption of active growth	Spring freeze damage to a crop that has broken dormancy and resumed active growth (more vulnerable to cold than the fully dormant plant); drought stress as the plant resumes water demand	USDA weekly Crop Progress resumes (condition ratings)	The first weekly condition ratings of the season, confirming or revising the winterkill/dormancy outlook
May	Heading and flowering	Drought and heat stress during heading; freeze damage to the now actively flowering crop; excess moisture promoting fungal disease pressure (Section 5)	USDA weekly Crop Progress (heading %, condition ratings)	A significant weather-sensitivity window, broadly analogous in agronomic role (though for a different crop and calendar) to corn's July pollination window
Jun–Jul	Harvest	Wet harvest conditions delaying fieldwork and degrading grain quality/test weight; harvest typically progresses from south (Texas/Oklahoma) to north as the season advances	USDA weekly Crop Progress (harvest %); July WASDE (first new-crop survey-based estimate)	Harvest pace and quality confirm or revise the production estimate; the July WASDE is the most closely watched single scheduled report of the wheat crop year

3.1 Winterkill — A Risk Unique to Winter Wheat Among Major US Row Crops

- Winter wheat must survive the winter as an established, dormant plant, exposing it to a weather risk — extreme cold without adequate insulating snow cover — that simply does not exist for spring-planted crops like corn and soybeans. Snow cover acts as an insulating layer; a cold snap arriving without snow on the ground (a "naked" cold event) is more damaging than the same temperature with several inches of snow present, making the combination of temperature and snow-cover data (rather than temperature alone) the relevant risk assessment during the dormancy window.

3.2 Spring Wheat — A Separate US Calendar

- Hard Red Spring wheat (priced via the Minneapolis contract, Section 2) follows a planting-to-harvest calendar similar in overall shape to corn's (spring planting, fully harvested within a single calendar year, no dormancy period), but shifted somewhat later given its more northerly Great Plains growing region. While not the subject of this CBOT/SRW-focused document, spring wheat's growing-season weather is occasionally relevant background context for the broader US wheat balance and for the cross-class spread relationships described in Section 2.1.

4. Global Production — A Genuinely Year-Round Global Harvest

Wheat is grown more widely across the globe, and across more offsetting hemispheres and harvest calendars, than any other major grain covered in this document series. The practical consequence is that there is, in effect, always an active wheat harvest occurring somewhere in the world, meaning the market's attention rotates through a sequence of regional harvests over the course of the year rather than concentrating on a single dominant window in the way corn's calendar concentrates on the US July pollination period.

Country/Region	Approx. Share of World Production	Harvest Period	Key Risk	Global Trade Role
European Union	~18–20%	Jul–Aug (Northern Hemisphere summer harvest)	Spring drought; harvest-time excess rainfall affecting quality	A major producer and exporter, particularly France and Germany; EU export policy and harvest size are closely tracked global supply variables.
China	~17–18%	Jun (winter wheat, the dominant Chinese class, harvested in the North China Plain)	Drought/heat in the North China Plain; domestic state reserve and import-policy management	The largest single producer but a relatively minor exporter, given large domestic consumption; Chinese import demand and reserve policy are tracked analogously to Chinese corn and cotton dynamics described elsewhere in this series.
India	~13–14%	Mar–Apr (Rabi/winter-season crop, harvested in spring)	Pre-harvest heat waves (a documented risk — see Section 8); irrigation water availability	A large producer, predominantly for domestic consumption; Indian government export policy (periodic restrictions to protect domestic food security) is a recurring market-moving variable.
Russia	~10–11%, the largest single exporter	Jul–Aug (winter wheat, the dominant Russian class)	Drought, particularly in the southern producing regions; winterkill in some years	The world's largest wheat exporter in most recent years; Russian export tax/quota policy and Black Sea logistics (Section 6) are central global wheat market variables.
United States	~6–7%	Jun–Jul (winter wheat dominant, per Section 3); spring wheat Aug–Sep	Winterkill; spring freeze; heading-stage drought/heat; harvest-time rain	A major exporter though a smaller share of global production and exports than several decades ago, reflecting the growth of Black Sea and other competing export origins.
Canada	~4–5%	Aug–Sep (predominantly spring wheat, given Canada's northerly growing regions)	Drought; early frost given the relatively short growing season at northern latitudes	A major exporter, particularly of high-protein spring wheat classes, competing directly with US HRS exports.

Ukraine	~2–3% in typical pre-war years, variable since 2022	Jul–Aug (winter wheat)	War-related disruption to planting, harvesting, and — most significantly — Black Sea export logistics since February 2022	A globally significant exporter whose Black Sea export access has been materially disrupted by the war; see Section 6.
Australia	~3–4%	Nov–Jan (Southern Hemisphere harvest)	Drought (El Nino years); excess rainfall/flooding damaging crop quality at harvest (La Nina years)	A major exporter, providing a Southern Hemisphere harvest pulse roughly six months out of phase with Northern Hemisphere production.
Argentina	~2–3%	Nov–Dec (Southern Hemisphere harvest)	Drought; export tax and export quota policy changes (the same recurring Argentine policy variable described for corn)	A significant Southern Hemisphere exporter; Argentine export policy is a periodically market-moving variable.

The combination of Northern Hemisphere summer harvests (US, EU, China, Russia, Ukraine, Canada, roughly May through September) and Southern Hemisphere summer harvests (Australia and Argentina, roughly November through January) means that, unlike corn's single concentrated US-centric weather-risk window, wheat's fundamental attention rotates through a near-continuous sequence of regional harvest and weather-risk windows across the full calendar year.

5. Disease and Quality Risk

Wheat quality — not merely yield — is a distinct and significant variable in the wheat market, since end-use applications (bread flour versus feed wheat, for example) depend on protein content, test weight, and the absence of disease-related grain damage, making disease pressure a relevant variable beyond its direct yield effect.

5.1 Fusarium Head Blight (Scab)

- A fungal disease that infects the wheat head during flowering under warm, humid conditions, reducing yield and — significantly — contaminating affected grain with vomitoxin (deoxynivalenol, DON), a mycotoxin subject to regulatory limits for both food and feed use. Severe Fusarium outbreaks can render a portion of the affected crop unsuitable for its intended milling use even where the physical yield loss is moderate, making this a quality-discount risk distinct from a pure yield-reduction risk.

5.2 Stripe Rust and Leaf Rust

- Fungal rust diseases (distinct from, but conceptually similar in disease mechanism to, the coffee leaf rust described elsewhere in this document series) that can spread rapidly under favourable humid conditions and reduce yield through damage to the plant's photosynthetic leaf tissue; resistant variety breeding and fungicide application are the primary management tools, and a significant regional outbreak is tracked by USDA and state extension services as a within-season risk factor.

5.3 Protein Content and Quality Premiums/Discounts

- Beyond disease-related quality issues, wheat protein content itself varies year to year with growing-season weather (generally, drier/hotter finishing conditions tend to concentrate protein, while a wetter finish tends to dilute it), and millers pay quality-based premiums and discounts in the cash market reflecting whether a given year's crop protein content meets the specifications required for bread-flour and other high-protein applications — a basis-level consideration layered on top of the futures price itself.

6. The Black Sea Region — Structural Importance to Global Wheat Trade

The Black Sea exporting region — principally Russia and Ukraine, with Romania and Kazakhstan as additional significant regional exporters — has grown over the past two decades into the most structurally important wheat export origin globally, making the region's logistics, weather, and (since 2022) the war in Ukraine central variables for the global wheat balance regardless of which specific futures contract a trader uses for price discovery.

6.1 Russia's Rise to the Largest Single Exporter

- Russia's wheat export volume has grown substantially since the 2000s, reflecting both production growth (driven by improved agronomic practices and the post-Soviet reorganisation of Russian and Ukrainian agriculture) and the country's geographic and logistical advantages (Black Sea port access providing relatively short and low-cost shipping routes to major importing regions in the Middle East, North Africa, and parts of Asia) relative to competing exporters.
- **Russian export tax and quota policy**: the Russian government has, at various points, implemented export taxes (calculated via a formula referencing the prevailing export price) and, at times, export quotas on wheat, explicitly intended to moderate domestic food price inflation by discouraging exports during periods of high global prices. Changes to this policy — whether tightening or loosening — are closely tracked, market-moving events given Russia's export market share.

6.2 The War in Ukraine and Black Sea Export Logistics

- Russia's full-scale invasion of Ukraine in February 2022 severely disrupted Ukrainian Black Sea port operations, given the direct conflict in and around the Black Sea and the blockade/contested-control conditions affecting Ukrainian export shipping. A series of subsequent arrangements and operational workarounds (including a formal grain export corridor agreement that operated for a period before its later suspension, and subsequent alternative shipping arrangements established by Ukraine along a coastal corridor) have evolved over time as the conflict has continued, with the specific operational status of Ukrainian Black Sea exports remaining subject to ongoing developments that traders should verify against current primary sources rather than treating any single point-in-time description as a stable, current fact.
- **The broader market effect**: regardless of the precise operational status at any given moment, the war has introduced a sustained layer of logistics and geopolitical-risk uncertainty into the global wheat balance that did not exist prior to February 2022, given the Black Sea region's combined (Russia plus Ukraine) share of global wheat exports.

7. Corn-Wheat Feed Substitution — The Primary Cross-Grain Relationship

Wheat and corn are both viable energy (caloric) components of livestock and poultry feed rations, and the relative price of the two grains determines, at the margin, which is used in feed formulation in regions and periods where both are economically accessible to feed manufacturers — making the corn-wheat price spread itself a closely tracked relative value relationship.

7.1 The Substitution Mechanism

- When wheat trades at a sufficiently low price relative to corn (on an energy/nutrient-equivalent basis), feed formulators substitute wheat — typically lower-grade or feed-quality wheat not meeting milling specifications — for a portion of the corn in livestock and poultry rations, increasing feed-wheat demand and providing a demand-side floor under wheat prices at low relative levels. When wheat is expensive relative to corn, this substitution demand recedes and feed formulators rely more heavily on corn.
- **Geographic and regulatory variation**: the prevalence of feed-wheat substitution varies significantly by region, reflecting differences in relative grain availability, milling industry by-product availability (which itself provides feed-grade material), and livestock industry structure; feed-wheat substitution is, for example, more structurally embedded in some European livestock feed markets than in the US market, where corn's overwhelming relative abundance in most years limits the economic incentive for wheat substitution except during periods of unusually wide price divergence.

7.2 The Corn-Wheat Spread as a Tracked Relationship

- The CBOT wheat-corn futures price spread (both quoted in ¢/bu, requiring no currency conversion since both are USD-denominated CBOT contracts) is tracked by feed-grain market participants as a direct read on the relative attractiveness of wheat-for-feed substitution at any given time, and is one of the most fundamentally grounded (rather than purely speculative) cross-commodity spreads in the agricultural futures complex, given the direct physical substitutability described above.

8. Global Demand and Food Security Policy

8.1 Wheat as a Staple Food Grain

- Wheat is among the most important direct human food staples globally (alongside rice and corn), consumed predominantly as bread, noodles, and other milled-flour products, making global wheat demand more directly tied to population growth and per-capita food consumption patterns than corn's demand structure, which (per the Corn reference document in this series) is substantially weighted toward feed and biofuel uses rather than direct human food consumption in the US specifically, though wheat too has a significant feed-use component as described in Section 7.

8.2 Government Food Security Policy and Export Restrictions

- Because wheat is such a central food-security staple in many countries, governments — particularly in major producing countries facing a domestic price spike or a poor harvest — have periodically implemented export restrictions or outright export bans specifically to protect domestic food supply and price stability, even at the cost of reduced export revenue. India's wheat export restriction implemented in 2022 (following a severe pre-harvest heat wave that reduced the expected crop, as referenced in Section 4) is a documented example of this policy mechanism in practice; similar restrictive measures have, at various points, been implemented or discussed by other major producing countries during periods of acute global price stress.
- **The market mechanism**: an export restriction by a major producer immediately removes that country's supply from the tradeable global pool (even though the wheat itself still physically exists, simply unavailable for export), which is mechanically equivalent in its effect on the tradeable global balance to an actual production shortfall of the same magnitude — making government policy announcements, not only weather and production data, a direct and immediate price-moving input for wheat.

8.3 Government and Institutional Tenders

- Many wheat-importing countries — particularly in the Middle East, North Africa, and parts of Asia — purchase a substantial share of their wheat import requirements via publicly announced government or state-trading-entity tenders (analogous in mechanism to the white sugar government tenders described elsewhere in this document series), making individual tender results (volume awarded, winning origin and price) a closely watched, relatively high-frequency global wheat demand signal.

9. The Forward Curve Structure — Zones and Spread Mechanics

The CBOT wheat curve's shape reflects the same general cost-of-carry framework as corn, modulated by the winter-wheat-specific calendar (autumn planting, winter dormancy, late-spring/early-summer harvest) described in Section 3, and by wheat's genuinely global, multi-hemisphere production calendar described in Section 4.

Contract	Crop-Year Role	Primary Driver	Secondary Driver	Key Signal
July (N)	New-crop anchor. The primary contract referencing the US winter wheat harvest just completed.	US winter wheat planted acreage (prior autumn); winterkill survival; heading-stage (May) weather; July WASDE first new-crop survey-based yield	Northern Hemisphere harvest progress more broadly (EU, Russia, Ukraine — all harvesting in a similar Jul-Aug window)	USDA weekly Crop Progress (heading, then harvest %); July WASDE; concurrent EU/Black Sea harvest reports
September (U)	New-crop, slightly deferred. US harvest essentially complete; export marketing underway.	US export sales pace vs USDA forecast; September WASDE/Grain Stocks (same-day release, as with corn)	Southern Hemisphere (Australia, Argentina) planting/early growing conditions, now entering their own growing season	Weekly export sales/inspections; September WASDE and Grain Stocks; early Southern Hemisphere crop condition reports
December (Z)	Old-crop, deferred. US winter wheat for the FOLLOWING year is being planted concurrently.	Autumn planting progress and conditions for the next US winter wheat crop; Southern Hemisphere harvest now underway and confirming that hemisphere's contribution to global supply	Russian/Black Sea export pace and any export policy changes (Section 6.1)	USDA weekly planting progress (next crop); Australian/Argentine harvest results; Russian export tax/quota news
March (H)	Old-crop, deep. Northern Hemisphere winter wheat in dormancy (the new crop, for harvest the following summer).	Winterkill risk monitoring for the crop now in dormancy; old-crop export pace and ending-stocks trajectory	Southern Hemisphere post-harvest marketing and export pace	Winter weather/snow-cover monitoring; weekly export data; Southern Hemisphere export shipment data
May (K)	Last old-crop contract before the new July anchor. US winter wheat entering its critical heading/flowering window.	US heading-stage weather (the most volatile window of the US wheat calendar, analogous in role to corn's July pollination); old-crop/new-crop (May-July) spread dynamics	Confirmed winterkill damage assessments now feeding into the yield outlook	Daily/weekly US temperature, precipitation, and freeze-risk monitoring during heading; weekly Crop Progress condition ratings

9.1 Why Wheat's Seasonal Pattern Is Less Singularly Weather-Driven Than Corn's

- Because wheat's global supply picture is assembled from a sequence of offsetting-hemisphere harvests throughout the year (Section 4) rather than concentrated around a single dominant US growing-season window, the CBOT wheat curve is, in relative terms, somewhat less purely dominated by US weather risk than the corn curve is by the US July pollination window — though US weather, and particularly the May heading-stage window and winter dormancy survival, remain significant and closely tracked variables for the SRW-specific physical market underlying CBOT futures.

10. Macro and Cross-Market Drivers

10.1 The Corn-Wheat Spread

- As detailed in Section 7, the corn-wheat feed substitution relationship makes this one of the most fundamentally grounded cross-commodity spreads in the agricultural complex, tracked continuously by feed-grain market participants.

10.2 US Dollar and Export Competitiveness

- As a globally traded, USD-denominated commodity with substantial export demand from a geographically diverse set of competing origins (Section 4), wheat exhibits the standard dollar-priced-commodity relationship described elsewhere in this document series: USD strength raises the effective cost of USD-denominated wheat exports for foreign buyers, a relative headwind for US (and other USD-referenced) export competitiveness against origins whose own currencies may be moving differently against the US dollar.

10.3 Russian Ruble (RUB/USD)

- Given Russia's position as the largest single wheat exporter (Section 6.1), RUB/USD movement affects Russian farmer and exporter economics through the same general mechanism described for BRL, VND, and other producer currencies elsewhere in this document series: RUB depreciation improves the rouble-equivalent revenue Russian exporters receive at a given USD wheat price, all else equal supporting Russian export competitiveness and volume, while RUB appreciation has the opposite effect.

10.4 Speculative Positioning — COT Data

- **CFTC COT for CBOT Wheat**: weekly managed money net positioning is closely tracked, consistent with the positioning framework described throughout this document series. Wheat has, at various points, been noted by market commentators for periods of unusually large or rapidly shifting speculative positioning relative to its underlying physical market size, a factor sometimes attributed to its frequent role as a vehicle for expressing broader geopolitical or food-security-related market views (e.g., around the Black Sea war's onset in 2022) beyond pure US-crop-specific fundamentals.

11. Documented Seasonal Patterns

Wheat's seasonal pattern, while still calendar-anchored around the US winter wheat cycle for the CBOT/SRW contract specifically, is layered with the additional complexity of the rotating global harvest sequence described in Section 4.

Period	US Crop Stage	Global Context	Typical ZW Behaviour	Reliability
Sep–Oct	Autumn planting of the new (next-July-harvest) US crop	Northern Hemisphere (EU, Russia, Ukraine) post-harvest export marketing in full swing	US planting conditions begin shaping the following year's new-crop outlook (December/March contracts); Black Sea export pace is the dominant near-term global driver	Moderate
Nov–Feb	Winter dormancy	Southern Hemisphere (Australia, Argentina) harvest underway and concluding	Generally lower US-specific weather attention; Southern Hemisphere harvest results and any winterkill cold-snap events are the primary catalysts	Lower — fewer high-frequency catalysts outside of acute cold events
Mar–Apr	Spring green-up; weekly Crop Progress resumes	Indian Rabi wheat harvest approaching (Mar-Apr) — a documented heat-wave risk window (Section 8.2)	First condition ratings of the season confirm or revise the winterkill outlook; Indian pre-harvest heat-wave headlines can move global wheat sentiment even though India is a minor exporter	Moderate, rising
May	Heading and flowering — the most weather-sensitive US window	EU and Black Sea crops also approaching their own critical growth stages on a similar calendar	The most volatile period of the US wheat weather calendar; overlapping critical-growth-stage risk across multiple Northern Hemisphere origins simultaneously can amplify a single adverse weather pattern's global market impact	High — the window itself is calendar-certain even though any specific weather outcome is not
Jun–Jul	US harvest	EU, Russia, and Ukraine harvests proceeding on a broadly similar Jul-Aug calendar	Harvest pace and quality (including Fusarium/quality-discount risk, Section 5) confirm or revise the US figure; July WASDE is the most closely watched scheduled report of the year	High
Aug–Sep	Post-harvest US marketing; spring wheat (a different US calendar) now also harvesting	EU/Black Sea harvest results fully confirmed; Southern Hemisphere planting underway for the upcoming Nov-Jan harvest	Export-pace data and the September WASDE/Grain Stocks data day are the primary near-term catalysts	Moderate

11.1 Documented Historical Events

- **2010 Russian drought and export ban**: a severe drought across Russian wheat-growing regions in summer 2010, combined with widespread wildfires, led the Russian government to implement a temporary export ban, removing a major supplier from the global market and contributing to a sharp global wheat price rally — a documented historical instance of the export-restriction mechanism described in Section 8.2.

- **2022 Russian invasion of Ukraine**: the war's onset in February 2022 produced an immediate, sharp spike in global wheat prices reflecting acute uncertainty over Black Sea export availability from two of the world's largest exporters simultaneously, before prices subsequently retreated as alternative shipping arrangements and other exporters' supply partially offset the disruption over the following months — illustrating both the acute initial shock and the global market's subsequent adjustment capacity described in Section 6.2.
- **2022 Indian export restriction**: following a severe pre-harvest heat wave that reduced the expected Indian wheat crop, India restricted wheat exports in May 2022, a documented instance of the food-security export-policy mechanism described in Section 8.2, occurring against the backdrop of the already-elevated global price environment following the Ukraine war's onset.

12. Key Data Sources and Release Schedule

Report	Publisher	Frequency	Typical Timing	Primary ZW Curve Impact
WASDE (Wheat component)	USDA WAOB	Monthly	~11th–12th of month, 12:00 ET	All contracts — monthly balance-sheet revision; July edition carries particular weight (first new-crop survey-based yield)
USDA Winter Wheat Seedings Report	USDA NASS	Annual	Mid-January	December/March (the following year's new crop) — first seeded-acreage signal for the crop now in/entering dormancy
USDA Prospective Plantings / Small Grains Summary	USDA NASS	Annual (Prospective Plantings); Sep (Small Grains Summary)	Late March; September	Acreage confirmation across all wheat classes and the full small-grains balance
USDA Weekly Crop Progress	USDA NASS	Weekly (resumes ~April through harvest)	Monday ~16:00 ET	Front contracts during the growing season — condition ratings through the May heading window are most watched
USDA Weekly Export Sales / Inspections	USDA FAS / AMS	Weekly	Thursday 08:30 ET (sales); Monday (inspections)	C1–C3 — export pace vs WASDE forecast; competition from Black Sea and Southern Hemisphere origins relevant context
Russian export tax/quota policy announcements	Russian Ministry of Agriculture	As scheduled/ amended	Variable	All contracts — given Russia's position as the largest single exporter
ABARES (Australian) and Argentine (Bolsa de Cereales) crop estimates	ABARES; Bolsa de Cereales de Buenos Aires	Seasonal/monthly	Variable	C3–C6 — Southern Hemisphere harvest confirmation, six months out of phase with Northern Hemisphere production
NOAA Drought Monitor and CPC temperature/precipitation outlooks	NOAA / NDMC / CPC	Weekly/continuous	Variable	Front contracts, acutely during winter dormancy (snow cover/cold) and the May heading window
International tender announcements (Egypt GASC and others)	Purchasing government/state entities; trade press	As scheduled, frequent	Variable	Global demand signal — individual tender results (volume, winning origin, price) are closely watched
CFTC Commitments of Traders	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding — all contracts; wheat periodically noted for unusually large positioning swings

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CBOT Wheat (Soft Red Winter) is the most globally distributed and multi-hemisphere-dependent of the major grain markets covered in this document series. Where corn's fundamental picture is overwhelmingly anchored to a single US growing season, wheat's global supply picture is assembled continuously across the calendar year from a sequence of Northern and Southern Hemisphere harvests, with the Black Sea region's structural importance — and the ongoing effects of the war in Ukraine on that region's export logistics — now a permanent fixture of the global wheat balance rather than a temporary disruption. Layered on top of this geographic complexity is wheat's multi-class structure (SRW, HRW, HRS, and others), each with its own dedicated futures contract and distinct end-use economics, and wheat's status as the most direct human-food-staple grain among the major row crops, which makes government food-security export policy a recurring, and sometimes acute, price-moving variable in a way that has a less direct parallel in corn's more feed-and-fuel-weighted demand structure.

For spread traders, the most reliable recurring framework is to track which of the rotating regional harvests (Section 4) is currently the market's dominant attention point at any given time of year, rather than assuming US-specific weather is always the primary driver in the way it reliably is for corn; to monitor the May-July old-crop/new-crop spread as the US-specific structural read, analogous in function to corn's July-December spread; to track Black Sea export policy and logistics as a now-permanent global risk factor; and to use the corn-wheat spread as the primary fundamentally-grounded cross-grain relative value signal connecting this market to the Corn reference document elsewhere in this series.